Lab 1 – Requirements Engineering & UML Use-Case Modelling

#09 – Faculty Research Grant & Publication Tracker

Use-Case Flow Specification

Use Case: Log Research Expenses

Primary Actor: Faculty Researcher

Supporting Use Case: Track Grant Budget

Preconditions

1. The Faculty Researcher is authorized to access the research grant system.

2. A valid research grant exists.

3. The grant has an available budget.

Postconditions

1. The research expense is recorded successfully.

2. The remaining grant balance is updated.

3. The expense is included in the grant's financial records.

Main Success Scenario

1. Faculty Researcher selects "Log Research Expenses".

2. The system displays the expense entry form.

3. Faculty Researcher enters the expense details, including amount, category, date, and description.

4. The system validates the entered information.

5. The system checks the available grant balance.

6. The system records the expense.

7. The system calculates the remaining grant budget.

8. The system displays a confirmation message and updated balance.

9. The use case ends successfully.

Alternate Flow – Expense Exceeds Available Grant Balance

1. The system detects that the expense exceeds the available grant balance.

2. The system rejects the expense.

3. The system displays an appropriate error message.

4. The Faculty Researcher can modify the expense amount or cancel the operation.

5. No invalid expense is recorded.